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로그 미분법 ($\frac{f'(x)}{f(x)} = (\ln|f(x)|)'$)
 $\rightarrow$ 함수의 꼴이 복잡하게 얽혀있을 때
 지수와 밑 모두 변할 때 $(f(x))^{g(x)}$ / x^x ...

$$f(x) = f(x) \cdot \{ \ln|f(x)| \}'$$

$$\left\{ 2\ln(x^2+2) + 4x + \ln(\sec(5\ln x)) - \frac{1}{2}\ln(x^2+1) \right\}'$$

$$= \frac{6x^2}{x^2+2} + 4 + \frac{-\sin(5\ln x)}{\sec(5\ln x)} \cdot 5\sec^2 x - \frac{1}{2} \cdot \frac{2x}{x^2+1}$$

$$f'(0) = f(0) \cdot (0 + 4 + 0 - 0) = 16$$

4.1.1

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$$\frac{\sin(\frac{\pi}{3})}{\frac{\pi}{3}} + \frac{\ln(\frac{\pi}{3})}{\frac{\pi}{3}} = -\frac{\pi}{9}$$

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$(\alpha, \beta) \Rightarrow y - \beta = -\alpha(x - \alpha)$

$$F: x^2y + \ln(1+xy) - \sec(y^2) - 3y = 0$$

$$\frac{dy}{dx} = - \frac{2xy + \frac{1}{1+\ln(xy)} \cdot \frac{y}{xy} + \sec^2(y^2) \cdot 2y}{x^2 + \frac{1}{1+\ln(xy)} \cdot \frac{x}{xy} + 2xy \cdot \sin(y^2) - 3} \Big|_{(0, -\frac{1}{3})} = -\frac{1}{9}$$

$$\therefore y = -\frac{1}{9}x - \frac{1}{3}$$

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$\frac{dx}{dt} = \frac{1}{x}$ $dx = x \cdot dt$ $x = \frac{1}{t}$ $dx = x \cdot dt$ $\ln e = 1$ $\ln 1 = 0$

$$\ln x^2 = t \Rightarrow dx = e^t dt \Rightarrow \int_0^1 t^2 e^t dt$$

$$2t = u \Rightarrow dt = \frac{1}{2} du$$

$$\frac{1}{2} \cdot \frac{1}{4} \int_0^2 u^2 e^u du = \frac{1}{8} [u^2 e^u - 2u e^u + 2e^u]_0^2 = \frac{1}{8} (4e^2 - 4e^2 + 2e^2 - 2) = \frac{1}{4} (e^2 - 1)$$

도표 작분법

$$[u^2 e^u - 2u e^u + 2e^u - 0 \cdot e^0]_0^2$$

$$= \frac{1}{8} (4e^2 - 4e^2 + 2e^2 - 2) = \frac{1}{4} (e^2 - 1)$$

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$$F = \frac{1}{x-0} \int_0^1 f(x) dx$$

x 의 범위: $0 \leq x \leq 1$
 t 의 범위: $x \leq t \leq 1$
 $0 \leq x \leq t \leq 1$

$$\int_0^1 \int_x^1 \cos(t^2) dt dx = \int_0^1 \int_0^t \cos(t^2) dx dt = \int_0^1 t \cos(t^2) dt = \left[\frac{1}{2} \sin(t^2) \right]_0^1 = \frac{1}{2} \sin 1$$

$\therefore F = \frac{1}{2} \sin 1$
 $\cos(t^2)$ 를 상수로 취급

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헤비사이드 정리
 $\frac{2}{x(x+1)} = \frac{A}{x} + \frac{B}{x+1}$ 에서 A, B 구하는 방법
 x 가 커지면, $x=0$ 대입 $\Rightarrow A=2$ (A=2)
 $x=-1$ 대입 $\Rightarrow B=-2$ (B=-2)
 $\frac{2}{x^2(x+1)^2}$ 의 경우엔?
 $= \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x+1} + \frac{D}{(x+1)^2}$ 에서 큰 차수만 처리

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(a) $\frac{1}{\pi} \int_{-\pi}^{\pi} \frac{e^{i\theta}}{1-e^{i\theta}} d\theta = \frac{1}{\pi} \int_{-\pi}^{\pi} \frac{1}{1-e^{i\theta}} d\theta$
 $\frac{1}{1-e^{i\theta}} = \frac{1}{1-\cos\theta - i\sin\theta} = \frac{1+\cos\theta + i\sin\theta}{(1-\cos\theta)^2 + \sin^2\theta} = \frac{1+\cos\theta + i\sin\theta}{2-2\cos\theta} = \frac{1+\cos\theta + i\sin\theta}{2(1-\cos\theta)}$
 $\frac{1}{1-\cos\theta} = \frac{1}{2} \frac{1}{\sin^2(\frac{\theta}{2})} = \frac{1}{2} \csc^2(\frac{\theta}{2})$
 $\frac{1}{\pi} \int_{-\pi}^{\pi} \frac{1}{2} \csc^2(\frac{\theta}{2}) d\theta = \frac{1}{2\pi} \int_{-\pi}^{\pi} \csc^2(\frac{\theta}{2}) d\theta = \frac{1}{2\pi} \int_{-\pi}^{\pi} -\cot(\frac{\theta}{2}) d\theta = \frac{1}{2\pi} \left[2 \ln|\sin(\frac{\theta}{2})| \right]_{-\pi}^{\pi} = 0$

(b) $\frac{1}{\pi} \int_{-\pi}^{\pi} \frac{1}{1-e^{i\theta}} d\theta = \frac{1}{\pi} \int_{-\pi}^{\pi} \frac{1}{1-\cos\theta - i\sin\theta} d\theta$
 $\frac{1}{1-\cos\theta - i\sin\theta} = \frac{1+\cos\theta + i\sin\theta}{(1-\cos\theta)^2 + \sin^2\theta} = \frac{1+\cos\theta + i\sin\theta}{2-2\cos\theta} = \frac{1+\cos\theta + i\sin\theta}{2(1-\cos\theta)}$
 $\frac{1}{1-\cos\theta} = \frac{1}{2} \frac{1}{\sin^2(\frac{\theta}{2})} = \frac{1}{2} \csc^2(\frac{\theta}{2})$
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(c) $\frac{1}{\pi} \int_{-\pi}^{\pi} \frac{1}{1-e^{i\theta}} d\theta = \frac{1}{\pi} \int_{-\pi}^{\pi} \frac{1}{1-\cos\theta - i\sin\theta} d\theta$
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 $\frac{1}{1-\cos\theta} = \frac{1}{2} \frac{1}{\sin^2(\frac{\theta}{2})} = \frac{1}{2} \csc^2(\frac{\theta}{2})$
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(가)에 의해

$$f(u) = u + u^2 + \sum_{n=3}^{\infty} \frac{f^{(n)}(0)}{n!} u^n = u + u^2 + \sum_{n=3}^{\infty} \frac{1}{n!} u^n = u + u^2 + \frac{1}{2} u^2 + \frac{1}{6} u^3 + \frac{1}{24} u^4 + \dots$$

$$= u + \frac{3}{2} u^2 + \frac{1}{6} u^3 + \frac{1}{24} u^4 + \dots$$

$$\frac{1}{n!} \sum_{k=3}^n \left(\frac{1}{k-2} - \frac{1}{k-1} \right) = \frac{1}{n!} \left(\frac{1}{1} - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \dots + \frac{1}{n-1} - \frac{1}{n} \right)$$

$$= \frac{1}{n!} \left(1 - \frac{1}{n} \right)$$

$$\therefore f(u) \leq 3 \leq m$$

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$$\left(\frac{1}{3} + 2\right)^n = \sum_{k=0}^n \binom{n}{k} \left(\frac{1}{3}\right)^k \cdot 2^{n-k}$$

$3^k \cdot 2^{n-k} \cdot 2^k$

$$r \leq n-1 \Rightarrow a_n = n a_{n-1} \cdot 3^{-n+1} \cdot 2^1 = 6n \left(\frac{1}{3}\right)^n$$

$$\therefore 6 \sum_{n=1}^{\infty} n \left(\frac{1}{3}\right)^n \cdot \frac{1}{3} \leq 1$$

$$\sum_{n=1}^{\infty} n r^n = 1 + 2r + 3r^2 + 4r^3 + \dots$$

$$\frac{1}{1-r} = 1 + r + r^2 + r^3 + \dots$$

$$\frac{1}{(1-r)^2} = 1 + 2r + 3r^2 + 4r^3 + \dots$$

$$\frac{1}{(1-r)^2} = 1 + 2r + 3r^2 + \dots \Big|_{r=\frac{1}{3}} = \frac{9}{4} \cdot \frac{1}{3} = \frac{3}{4}$$

$$\therefore 6 \cdot \frac{3}{4} = \frac{9}{2}$$

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(sol I)
<일반좌표계>

$$\vec{AC} = (a, 0, a), \vec{AD} = (a, a, a)$$

$$\vec{AB} \times (\vec{AC} \times \vec{AD}) = (-a^3, a^3, 0) \Rightarrow \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right)$$

$$|\vec{AC}| = 1 \Rightarrow a = \frac{1}{\sqrt{2}}$$

$$\therefore (=) \frac{1}{\sqrt{2}} \vec{DC}$$

(sol II)

$$\vec{AB} \times (\vec{AC} \times \vec{AD}) = (\vec{AB} \cdot \vec{AD}) \vec{AC} - (\vec{AB} \cdot \vec{AC}) \vec{AD}$$

$$|\vec{AB}| |\vec{AD}| \cos \frac{\pi}{3} = \frac{1}{2} \quad |\vec{AB}| |\vec{AC}| \cos \frac{\pi}{3} = \frac{1}{2}$$

$$= \frac{1}{\sqrt{2}} (\vec{AC} - \vec{AD}) = \frac{1}{\sqrt{2}} \vec{DC}$$

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$$x^2 y^2 = 64y$$

$$r^2 \cos^2 \theta + r^2 \sin^2 \theta = 64 \cos^2 \theta$$

$$\Rightarrow r = \frac{64 \cos^2 \theta}{1 + \sin^2 \theta}$$

$$\text{이때, } \cos \theta = t \Rightarrow r = \frac{64(1-t^2)}{1+t^2}$$

$$A = \frac{1}{2} \int_0^{\pi} r^2 d\theta \xrightarrow{\text{대입}} \frac{1}{2} \int_0^{\pi} \frac{36t^2(1+t^2)}{(1+t^2)^2} \cdot \frac{1}{1+t^2} dt$$

$$= 18 \int_0^{\infty} \frac{t^2}{(1+t^2)^2} dt \xrightarrow{t^2=u} 6 \int_0^{\infty} \frac{1}{(1+u)^2} du = 6$$

* 세카르트 변환 *

$$x^2 + y^2 = 32xy \Rightarrow \text{변환: } \frac{y}{x} = t$$

$$\begin{cases} x = \frac{32t}{1+t^2} \\ y = \frac{32t^2}{1+t^2} \end{cases} \quad (d\theta = \frac{1}{1+t^2} dt)$$

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$$af = \left\langle -\frac{1}{2}, -\frac{1}{2}, 1 \right\rangle \Big|_{(a,b,c)} = \left\langle -\frac{a}{2}, -\frac{b}{2}, 1 \right\rangle \parallel \langle 1, 1, -1 \rangle$$

$$\therefore a=2, b=2, c=2 \Rightarrow a+b+c=6$$

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$$\text{점평면: } z = \frac{z_1(3,5)(x-3) + z_2(3,5)(y-5) + 2}{\frac{3}{2} - \frac{1}{2}}$$

$$x=3.03, y=4.98 \Rightarrow z \approx 2.05$$

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$$g: x^2 - x^2 + y^2 - y^2 = 0 \quad \& \quad f = y$$

$$\left| \frac{f}{g} \right| = \left| \frac{0}{x^2 - x^2 + y^2 - y^2} \right| = (x^2 - y^2) = 0 \quad \therefore x=0 \text{ or } x=\pm \frac{1}{\sqrt{2}}$$

$$\therefore x=0: y^2(y^2-1)=0 \Rightarrow y=-1, 0, 1$$

$$\therefore x=\pm \frac{1}{\sqrt{2}}: y^2 - y^2 - \frac{1}{4} = 0 \Rightarrow y^2 = \frac{1+\sqrt{2}}{2} (\because y^2 > 0)$$

$$\therefore y = \pm \sqrt{\frac{1+\sqrt{2}}{2}}$$

$$\therefore M = \sqrt{\frac{1+\sqrt{2}}{2}}, \quad m = -\sqrt{\frac{1+\sqrt{2}}{2}} \Rightarrow M \cdot m = -\frac{1+\sqrt{2}}{2}$$

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(sol I)

$$f_y = \frac{1}{1+(xy)^2} + 4048xye^{y^2-1}$$

$$\int_0^1 \int_0^1 \frac{f_y}{xy} dx dy = \int_0^1 \left[\frac{1}{1+(xy)^2} + 4048xye^{y^2-1} \right]_{x=0}^1 dy$$

$$\frac{1}{1+y^2} - 4048ye^{y^2-1}$$

$$\Rightarrow \left[\arctan y + 2024e^{y^2-1} \right]_0^1 = \frac{\pi}{4} + 2024 - \frac{2024}{e}$$

(sol II)

$$\int_0^1 \int_0^1 f_{xy}(x,y) dx dy$$

$$[f_y(x,y)]_0^1 = f_y(1,y) - f_y(0,y)$$

$$= [f(1,y)]_0^1 - [f(0,y)]_0^1 = f(1,1) - f(1,0) + f(0,1) - f(0,0)$$

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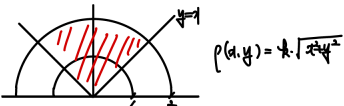
$$x^2+y^2=1, y \geq 0 \Rightarrow J(x,y)=1 \Rightarrow J(u,v)=1.$$

$$\Rightarrow \int_0^1 e^{u^2} du$$

$$\text{변환} \rightarrow (0,0), (1,0), (1,1)$$

$$\Rightarrow \int_0^1 \int_0^u e^{u^2} du dv = \int_0^1 u e^{u^2} du = \left[\frac{1}{2} e^{u^2} \right]_0^1 = \frac{1}{2}(e-1)$$

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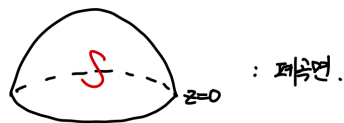


$$\bar{y} = \frac{M_y}{m}$$

$$m = \int_0^{\pi/2} \int_0^1 r^2 dr d\theta = \frac{\pi}{2} \cdot \frac{1}{3} \cdot 2 = \frac{2\pi}{3}$$

$$M_y = \int_0^{\pi/2} \int_0^1 r^3 \sin\theta dr d\theta = \left[-\cos\theta \right]_0^{\pi/2} \cdot \frac{1}{4} (1-0) = \frac{1}{4} \left(\frac{1}{2} + \frac{1}{2} \right) \Rightarrow \bar{y} = \frac{4\sqrt{2}}{16\pi}$$

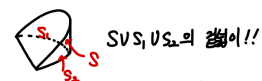
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$$\iint_S \mathbf{F} \cdot \mathbf{n} dS \stackrel{\text{Divergence}}{=} \iiint_V \underbrace{\text{div} \mathbf{F}}_{22} dV = 2 \int_0^{2\pi} \int_0^{\pi/2} \int_0^1 \rho \cos\theta \cdot \rho^2 \sin\theta d\rho d\theta d\phi$$

$$= 4\pi \cdot \left[\frac{1}{2} \sin^2\theta \right]_0^{\pi/2} \cdot \left[\frac{1}{4} \rho^4 \right]_0^1 = 2\pi \cdot 1 = 2\pi$$

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$$S_2: x^2+y^2 \leq 1, z=0: \frac{\pi}{2}$$

$$S_1: \int \sqrt{1+z} dA = 2 \int_0^1 \int_0^{2\pi} \sqrt{1+z} r dr d\theta = \pi$$

$$S: \mathbf{r}(\theta, z) = (\cos\theta, \sin\theta, z) \quad (0 \leq \theta \leq 2\pi, 0 \leq z \leq \sqrt{1-\cos^2\theta})$$

$$= \int_0^{2\pi} \int_0^{\sqrt{1-\cos^2\theta}} |\mathbf{r}_\theta \times \mathbf{r}_z| dz d\theta = 2\sqrt{3}$$

$$\therefore \text{합} = \frac{3}{2}\pi + 2\sqrt{3}$$

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$$\mathbf{r} = \langle 2\cos\theta\cos\phi, 2\cos\theta\sin\phi, 2\sin\theta \rangle \quad (0 \leq \theta \leq \frac{\pi}{2}, 0 \leq \phi \leq 2\pi)$$

$$\int_0^{2\pi} \int_0^{\pi/2} 4\cos^2\theta \cdot \underbrace{|\mathbf{r}_\theta \times \mathbf{r}_\phi|}_{4\sin\theta} d\phi d\theta = 16 \cdot 2\pi \left[-\frac{1}{3} \cos^3\theta \right]_0^{\pi/2} = \frac{32\pi}{3}$$